

Why is $\sin 2x = 2 \sin x \cos x$?

This is the double-angle formula, ~~not~~ derived from the sum of angles formula:

Derivation:

We know that:

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

if we set $A = B = x$:

$$\sin(x+x) = \sin x \cos x + \cos x \sin x$$

$$\sin(2x) = 2(\sin x)(\cos x) = 2 \sin x \cos x$$

